

Lab 16 Backtracking

Partea I

1. Folosind backtracking afișați toate permutările unui set de numere dat.
2. Folosind backtracking afișați toate submulțimile unui set de numere dat.
3. Folosind backtracking implementați problema reginelor pentru o tablă de dimensiune 4*4.

Partea II

1. Folosind backtracking implementați problema comis voiajorului.
2. [Numere de telefon](#)
3. [Adrese IP](#)
4. [SUDOKU](#)

Rezultate așteptate

[outputPermutari.txt](#)

```
0 2 4
0 4 2
2 0 4
2 4 0
4 0 2
4 2 0
```

[outputSubmultimi.txt](#)

```
8
4
4 8
2
2 8
2 4
2 4 8
0
0 8
0 4
0 4 8
0 2
0 2 8
0 2 4
0 2 4 8
```

[outputRegine.txt](#)

1	3	0	2
2	0	3	1

Schelet backtracking

[backtracking.c](#)

```
#define _CRT_SECURE_NO_WARNINGS
#include <conio.h>
#include <stdlib.h>
#include <stdio.h>
#include <string.h>
#include <math.h>

#define N 3

//PERMUTARI
typedef struct partialSolution {
    int originalVector[N] = { 0, 2, 4 };
    int permutation[N] = { -1, -1, -1 };
    int position = -1;
    int step = -1;
}partialSolution;

//SUBMULTIMI
typedef struct partialSolution {
    int originalVector[N] = { 0, 2, 4, 8 };
    int selector[N] = { -1, -1, -1, -1 };
    int position = -1;
    int step = -1;
}partialSolution;

//REGINE
typedef struct partialSolution {
    int queens[N] = { -1, -1, -1, -1 };
    int position = -1;
    int step = -1;
}partialSolution;

int canReject(partialSolution PS) {
}

int isSolution(partialSolution PS) {
}

void printSolution(partialSolution PS) {
}
```

```
partialSolution increaseStep(partialSolution PS) {  
}  
  
partialSolution getNextChoiceAtStep(partialSolution PS) {  
}  
  
int hasChoiceAtStep(partialSolution PS) {  
}  
  
void backtracking(partialSolution PS) {  
    if (canReject(PS))  
        return;  
    if (isSolution(PS))  
        printSolution(PS);  
    PS = increaseStep(PS);  
    while (hasChoiceAtStep(PS)) {  
        PS = getNextChoiceAtStep(PS);  
        backtracking(PS);  
    }  
}  
  
int main()  
{  
    partialSolution initialValues;  
    backtracking(initialValues);  
    return 0;  
}
```

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<https://wiki.mta.ro/> - **Cursuri Academia Tehnică Militară "Ferdinand I"**

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Last update: **2024/05/07 14:28**